

RENYUAN LIU

+86 14784206312 ◇ renyvannnn@gmail.com

EDUCATION

- **The Chinese University of Hong Kong, Shenzhen** Sept. 2026 – Jun. 2028 (Expected)
MPhil in Artificial Intelligence; Supervisor: Prof. Junjie Hu
- Guangzhou University Sept. 2022 – Jun. 2026 (Expected)
B.Eng. in Information Security; GPA: **89.81**/100.00; Ranking: **Top 8%**
Curriculum: Machine Learning 100, Operating System 98, Data Structure and Algorithm 97.

PAPERS

- **R. Liu** and Q. Fu, [Attention-Driven LPLC2 Neural Ensemble Model for Multi-Target Looming Detection and Localization](#). *The 2025 International Joint Conference on Neural Networks (IJCNN)*(CCF-C, AR $\approx 38\%$).
- G. Gao^{*}, **R. Liu**, M. Wang and Q. Fu^{*}, [A Computationally Efficient Neuronal Model for Collision Detection with Contrast Polarity-Specific Feed-Forward Inhibition](#). *Biomimetics*, vol.9, no.11, p.650, 2024 (JCR Q1, IF = 3.9).
- C. Fang^{*}, H. Zhou, **R. Liu**, and Q. Fu^{*}, [A Neuromorphic Binocular Framework Fusing Directional and Depth Motion Cues Towards Precise Collision Prediction](#). *Neurocomputing*, 131660 (JCR Q1, CCF-C, IF = 6.5).
- H. Zhou, C. Fang, **R. Liu**, and Q. Fu, [A Bio-Plausible Neural Network Integrating Motion and Disparity Pathways for Looming Perception](#). *Acta Electronica Sinica*, p.1-16, 2025 (CCF-A in Chinese Category).
- J. Huang^{*}, Z. Qin, M. Wang, **R. Liu**, and Q. Fu^{*}, [A Biomimetic Collision Detection Visual Neural Model Coordinating Self-and-Lateral Inhibitions](#). *The 14th International Conference on Biomimetic and Biohybrid Systems (Oral)*.
- **R. Liu**, H. Zhou, C. Fang and Q. Fu, [How Fly Neural Perception Mechanisms Enhance Visuomotor Control of Micro Robots](#). *Under review in 2026 International Conference on Robotics and Automation (ICRA)*(CCF-B).
- M. Wang^{*}, **R. Liu**^{*} and Q. Fu, [Enhancing Collision-Selectivity in Autonomous Micro-Robots by Elevated Temporal Derivatives in Neuronal Assembly Framework](#). *Under review in IEEE Transactions on Cognitive and Developmental Systems (JCR Q1, IF = 4.9)*.

HONORS AND AWARDS

- **First Prize (Provincial; Top 3%)**, Chinese Collegiate Computing Competition (4C) May 2025
- **Honorable Mention (International)**, Mathematical Contest in Modeling (MCM) Jan. 2025
- **First Prize (National; Top 5%)**, Asia and Pacific Mathematical Contest in Modeling (APMCM) Nov. 2024
- **First Prize & Innovation Silver Award (Provincial; Top 2 out of 1,167 Teams)**,
“Greater Bay Area Cup” Guangdong-Hong Kong-Macao Financial Mathematics Modeling Competition Nov. 2024
- **Commendation Letter** for Outstanding Performance in the Winning Team,
Interdisciplinary Programme at University of Hong Kong and The University of Macao Nov. 2023
- **Second; Third; First-Class Scholarship (Top 8; 12; 5%)**, *Guangzhou University* Dec. 2025; 2024; 2023

SKILLS

- **Coding:** C/C++, Python, Matlab, TypeScript **English:** IELTS 6.5, CET-6 564
- **Skills:** Claude Code/Codex/OpenCode, Transformer, Ascend 910B, Agent skills, Docker, FastAPI, PyTorch, Hugging Face, Git, ROS, STM32, Webots, Keil, MS Office, MS Visio, Adobe PS/Pr

INTERNSHIP EXPERIENCE

Shenzhen Research Institute of Big Data (SRIBD), Shenzhen, China

Role: **Algorithm Group Intern** Advisors: Jianghua Wu and Akang Wang

Mar. 2026 – Present

- **AI agent application development:** Developed a web-based AI education platform from scratch, including agent-driven news screening, generation and push modules, learning modules, and a coding agent creative workshop; also developed the supporting Android application.
- **Large model algorithms:** Deployed Qwen models on the **Ascend 910B** accelerator and optimized inference speed; developed an LLM routing system whose semantic-based algorithm reduced cost by **62%** while controlling the accuracy loss within **10%**; currently continuing work on large-model deployment and optimization.

RESEARCH EXPERIENCE

Computational Autonomous Learning Systems Lab Advisor: **Prof. Pengcheng Liu**

Department of Computer Science, University of York, York, UK (On-Site)

- **Biologically Plausible Mechanisms for Long-Term Motion Learning.** Jun. 2025 – Sept. 2025
 - **Robotic arm motion planning:** Learned expert-guided trajectory optimization via Learning from Demonstration (**LfD**) and applied biologically inspired probabilistic movement primitives (**ProMPs**) for push-grasping with the *Franka Emika Panda*.
 - **Navigation and manipulation:** Working on developing a **lifelong learning** navigation-manipulation system on *TurtleBot 3* with *OpenMANIPULATOR-X* that adapts to new environments while **retaining** performance in previously learned ones.

Machine Life and Intelligence Research Centre Advisor: **Prof. Qinbing Fu**

School of Mathematics and Information Sciences, Guangzhou University, Guangzhou, China

- **Real-time Visual Processing Systems Development of Micro-Mobile Robot** Mar. 2023 – Present
 - Developed the multi-attention LPLC2 (mLPLC2) neural network model inspired by the visual system of the fly by leveraging a **bottom-up attention** mechanism driven by motion-sensitive neural pathways (**independently, 3k lines of code in C/C++**).
 - Deployed visual neural network models inspired by insect neurons onto the **STM32-based micro-robot *Colias***, achieving real-time collision perception and avoidance. Optimized model memory usage to fit within the **62 KByte** SRAM capacity of *Colias*; developed and refined algorithms to enable real-time execution under extreme computational constraints (processing time < **33 ms** on the STM32F427 micro chip); performed debugging, tuning, and conducted both offline and online experiments.
 - **Fly-Inspired Ultra-selective Looming Perception and Avoidance on Resource-Constrained Micro-Robots**, poster presentation at *the 26th Towards Autonomous Robotic Systems (TAROS 2025) Conference*.
 - Selected code can be accessed below:
Fly Visuomotor-Inspired Attention-LPLC2 Model (**independently, 2k lines of code in C**);
Locust Vision-Inspired Optimized-LGMD Model (**independently, 1k lines of code in C**).